Height-Aware Attenuation Priors for Dense UAV Channel Knowledge Maps

Guillem Moreno Garcia, Evgenii Vinogradov, Sergi Abadal
Universitat Politecnica de Catalunya (UPC), Barcelona, Spain
guillem.moreno.garcia@estudiantat.upc.edu

***Abstract*—Dense channel knowledge maps can replace repeated ray tracing in radio planning, but a useful surrogate must preserve the strong height, range, and visibility structure of air-to-ground attenuation. This paper presents a height-aware, auditable attenuation prior for dense unmanned aerial vehicle channel knowledge maps. A line-of-sight branch combines free-space loss with a coherent image-source two-ray term and training-only height/range calibration. A non-line-of-sight branch combines established urban and air-to-ground trends with local morphology descriptors, followed by regime-specific linear calibration. A geometric visibility mask selects the deployed prior. The evaluation uses a strict city holdout with 2590 maps from 14 unseen cities, each containing 513×513 receiver locations at 7.125 GHz. The prior reaches 1.9383 dB overall root mean square error, with 1.7519 dB in line of sight and 3.5143 dB in non-line of sight, while attaining a per-map spatial correlation of 0.9477. The result provides a strong, inspectable starting point for fast UAV radio-map generation and for later residual learning.**

***Index Terms*—air-to-ground channel, channel attenuation, channel knowledge map, path loss, radio map, unmanned aerial vehicle.**

I. Introduction

Channel knowledge maps (CKMs) organize location-dependent channel information so that planning and adaptation do not require a new measurement or ray-tracing query for every link [1], [2]. The idea is especially relevant to unmanned aerial vehicle (UAV) networks, where transmitter height changes visibility, propagation distance, and the interaction with the urban skyline [3], [4]. A dense CKM surrogate must therefore reproduce more than a plausible image: it must preserve radial attenuation structure, line-of-sight (LoS) and non-line-of-sight (NLoS) regimes, and their dependence on height.

Data-driven radio-map methods can achieve accurate dense predictions [5], but an unconstrained regressor may spend capacity recovering simple propagation structure and may hide a weak NLoS regime behind a global metric. Purely analytical models present the opposite problem. Free-space and two-ray models capture range and coherent reflection, while classical urban and air-to-ground (A2G) models describe broad blocked-link trends [6], [7]. Their original coefficients and deployment domains, however, do not directly reproduce the high-resolution ray-traced CKM maps considered here.

The state-of-the-art limitation addressed in this work is the absence of a single, inspectable attenuation anchor that combines these two views under a strict unseen-city protocol. The proposed prior keeps literature-derived propagation structure, but fits all CKM-specific coefficients using training cities only. LoS pixels use a coherent two-ray model with smooth height and radial calibration. NLoS pixels use a conservative urban/A2G envelope plus local morphology features. A geometric visibility mask combines both maps.

The main contributions are:

- a height-aware LoS prior that preserves coherent radial rings instead of reducing the link to distance-only loss;
- a morphology-aware NLoS prior whose raw propagation estimate, local building descriptors, and fitted regime remain separately inspectable; and
- a leakage-resistant evaluation on 14 unseen cities, including overall, LoS/NLoS, height, and map-structure diagnostics.

Unlike a later multi-target residual model, this paper is deliberately limited to the attenuation prior. Delay spread, angular spread, and neural residual learning are outside its contribution.

A. Related Work and Positioning

CKMs describe location-indexed channel side information rather than a single link budget. Their uses include environment-aware scheduling, beam selection, and trajectory or deployment planning [1], [2]. Dense attenuation-map prediction is commonly approached as image-to-image regression. RadioUNet, PMNet, and related challenge systems demonstrate the strength of convolutional encoder–decoder models when topology and sparse or simulated radio information are available [5], [8]–[10]. These methods motivate dense output, but their evaluation settings do not isolate a continuously varying elevated transmitter under a strict city holdout.

Hybrid models address the complementary weakness of pure image regression: known propagation structure should not have to be rediscovered independently in every city. Free-space and coherent two-ray models explain the dominant range, height, and reflection structure of unobstructed links [11]–[13]. COST231/Hata and air-to-ground models provide coarse blocked-link and elevation trends [3], [6], [7]. However, their nominal coefficients and validity ranges cannot simply be transferred to a dense 7.125 GHz ray-traced dataset. We therefore retain their functional structure as a prior and fit only compact calibration terms on training cities.

This positioning differs from learning a correction to a single path-loss formula. The LoS branch explicitly preserves coherent rings, whereas the NLoS branch combines a conservative analytical scale with observed local morphology. It also differs from probabilistic LoS models: a deterministic visibility mask is available here and is used to route each receiver through the appropriate branch. The result is a fully dense attenuation estimate whose intermediate quantities remain accessible for diagnosis.

II. Data and Evaluation Contract

Each sample is a 513×513 urban CKM with $1\,\mathrm{m}$ spatial resolution. One elevated transmitter is horizontally centered in the map, operates at $7.125\,\mathrm{GHz}$, and has a sample-dependent height $h_{\mathrm{tx}} \approx 12\,\mathrm{m}$ to $478\,\mathrm{m}$. Every non-building pixel is a valid receiver at $h_{\mathrm{rx}} = 1.5\,\mathrm{m}$; building pixels are excluded from the metric rather than counted as zero-error targets. The prediction target is the stored dense channel-attenuation map in decibels.

The split is made by city rather than by pixel, tile, or random sample. The frozen attenuation artifact was calibrated on 11185 maps from 38 cities. A further 2405 maps from 14 disjoint cities form the development partition, and the final test set contains 2590 maps from 14 further cities. The calibration partition is the 37-city thesis training set plus Fortaleza; correspondingly, the development set is the thesis validation set without Fortaleza. No final test city occurs in either coefficient fitting or lookup-table construction. All reported claims use the final test partition after the fitted quantities have been frozen.

For valid-receiver mask $m_i(p)$, prediction $\hat{y}_i(p)$, and target $y_i(p)$, the pixel-weighted test error is

$$\mathrm{RMSE} = \sqrt{\frac{\sum_i \sum_p m_i(p)[\hat{y}_i(p) - y_i(p)]^2}{\sum_i \sum_p m_i(p)}}. \tag{1}$$

We also report mean absolute error (MAE) and map correlation. Let $\Omega_i = \{p : m_i(p) = 1\}$, $N_i = |\Omega_i|$, and let $\widetilde{\mathbf{y}}_i$ and $\widetilde{\hat{\mathbf{y}}}_i$ denote the valid target and prediction vectors after subtracting their respective per-map means. Then

$$\begin{aligned} \rho_i &= \frac{\widetilde{\hat{\mathbf{y}}}_i^{\top} \widetilde{\mathbf{y}}_i}{\|\widetilde{\hat{\mathbf{y}}}_i\|_2 \|\widetilde{\mathbf{y}}_i\|_2}, \\ \mathrm{MapCorr} &= \frac{\sum_{i \in \mathcal{C}} N_i \rho_i}{\sum_{i \in \mathcal{C}} N_i}. \end{aligned} \tag{2}$$

The set $\mathcal{C}$ excludes maps with fewer than two valid pixels or zero variance in either vector. Thus, map correlation is the valid-pixel-weighted mean of per-map Pearson correlations, not one correlation after concatenating all cities. RMSE remains the primary unit-aware metric; map correlation checks whether relative high- and low-attenuation regions are placed correctly.

III. Height-Aware Attenuation Prior

Fig. 1 summarizes calibration and deployment. The method uses known geometry to compute distances, a building/visibility mask to route LoS and NLoS pixels, and local

TABLE I
EVALUATION SETTING.

Item	Setting
Map	513×513, $1\,\mathrm{m}$ pixels
Carrier	$7.125\,\mathrm{GHz}$
Transmitter	Centered; variable height
Receiver	Ground, $1.5\,\mathrm{m}$; buildings masked
Split	11185/2405/2590 calibration/development/test maps by city
Test cities	Barcelona, Beijing, Carcassonne, Copenhagen, Halifax, Jaipur, Johannesburg, Key West, Kuala Lumpur, Osaka, Phuket, Pune, Surat, Vancouver

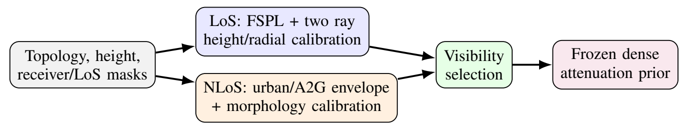

Fig. 1. The prior keeps LoS and NLoS propagation mechanisms separate. All CKM-specific calibration is fitted on training cities and then frozen.

windows over the building-height map. Only compact calibration tables and linear weights are learned. They are frozen before a held-out city is evaluated.

A. LoS Branch

Let d_{2D} be the horizontal transmitter-receiver distance. The direct and image-source reflected distances are

$$\begin{aligned} d_{\mathrm{LoS}} &= \sqrt{d_{2D}^2 + (h_{\mathrm{tx}} - h_{\mathrm{rx}})^2}, \\ d_{\mathrm{ref}} &= \sqrt{d_{2D}^2 + (h_{\mathrm{tx}} + h_{\mathrm{rx}})^2}. \end{aligned} \tag{3}$$

The free-space term follows the Friis distance dependence, while the coherent correction retains the phase difference between the direct and reflected paths [11]–[13]:

$$\mathrm{FSPL} = 32.45 + 20\log_{10}(d_{\mathrm{LoS}}/1000) + 20\log_{10}(f_{\mathrm{MHz}}), \tag{4}$$

$$C_{2\mathrm{ray}} = -20\log_{10}\left|1 + \rho(h_{\mathrm{tx}})\frac{d_{\mathrm{LoS}}}{d_{\mathrm{ref}}} e^{-j[k(d_{\mathrm{ref}} - d_{\mathrm{LoS}}) + \phi(h_{\mathrm{tx}})]}\right|. \tag{5}$$

Here ρ is an effective reflected-field amplitude and ϕ is a fitted phase offset. They are CKM calibration terms, not material Fresnel coefficients. The deployed LoS prior is

$$\mathrm{PL}_{\mathrm{LoS}} = \mathrm{FSPL} + C_{2\mathrm{ray}} + \beta_{\mathrm{LoS}}(h_{\mathrm{tx}}) + r(d_{2D}, h_{\mathrm{tx}}). \tag{6}$$

Calibration uses valid LoS pixels from calibration-side cities only. In each $5\,\mathrm{m}$ height bin, a small grid search selects effective amplitude and phase. A closed-form bias removes the remaining vertical offset. Finally, a smoothed radial table stores the mean residual left by the coherent model. Neighboring height bins are smoothed and linearly interpolated at inference. This structured sequence targets the constructive/destructive rings visible around the centered transmitter without giving the prior enough flexibility to memorize building layouts.

In implementation terms, the LoS correction is a lookup table. The stored entries are the effective ρ, ϕ, and bias curves indexed by transmitter-height bin, together with r indexed by height and radial-distance bin. Inference linearly interpolates neighboring height entries and reads the radial correction for each receiver; it does not refit on the new map. This distinction matters: although the table is data-calibrated, all 14 reported test cities are unseen during table construction. The free-space/two-ray structure supplies extrapolatable geometry, while the lookup table supplies a compact correction learned on other environments.

For each height bin, candidates $\hat{\rho} \in \{0, 0.05, \ldots, 1.5\}$ and 48 uniformly spaced phases in $[-\pi, \pi)$ are evaluated. For a candidate pair, the remaining bias is

$$\hat{\beta}(\hat{\rho}, \hat{\phi}) = \frac{1}{|B_h|} \sum_{i \in B_h} \left[y_i - \mathrm{FSPL}_i - C_{2\mathrm{ray},i}(\hat{\rho}, \hat{\phi}) \right]. \quad (7)$$

The pair minimizing the bin RMSE is retained. After smoothing the height curves, the remaining error is averaged by radius and stored as r. This ordering prevents the table from encoding arbitrary two-dimensional building texture.

B. NLoS Branch

The blocked-link branch starts from two coarse shapes. The first is a COST231/Hata-derived urban trend; the second is an elevation-dependent A2G NLoS envelope [3], [6], [7]. Their original domains do not directly match this CKM setup, so they are used as basis functions rather than transferred as final laws. Each map has 513×513 cells at 1 m spacing, with the transmitter projection at cell $(256, 256)$. For a receiver cell $p = (u, v)$, the implementation first computes

$$\begin{aligned} d_{2D}(p) &= 1\,\mathrm{m}\sqrt{(u-256)^2 + (v-256)^2}, \\ d(p) &= \max\{d_{2D}(p), 1\,\mathrm{m}\}, \\ h_t &= \max\{h_{\mathrm{tx}}, 1\,\mathrm{m}\}, \quad h_r = \max\{h_{\mathrm{rx}}, 0.5\,\mathrm{m}\} = 1.5\,\mathrm{m}. \end{aligned} \quad (8)$$

Thus $d_{\mathrm{km}} = \max\{d(p)/1000\,\mathrm{m}, 0.001\}$; all logarithms below receive the numerical value in the stated unit. With $a(h_{\mathrm{rx}}) = (1.1\log_{10} f_{\mathrm{MHz}} - 0.7)h_{\mathrm{rx}} - (1.56\log_{10} f_{\mathrm{MHz}} - 0.8)$, and $\eta(h_t) = 44.9 - 6.55\log_{10} h_t$, the first basis is

$$\begin{aligned} \mathrm{PL}_C = 46.3 + 33.9\log_{10} f_{\mathrm{MHz}} - 13.82\log_{10} h_t - a(h_r) \\ + \eta(h_t)\log_{10} d_{\mathrm{km}} + 3. \end{aligned} \quad (9)$$

The $+3$ dB metropolitan correction is retained inside this empirical basis; Eq. (9) is not claimed to be a directly valid COST231 deployment law at the carrier frequency and heights considered here. At $f = 7.125\,\mathrm{GHz}$, let $\theta(p) = (180/\pi)\tan^{-1}[(h_t - h_r)/d(p)]$ and $\lambda_0 = 20\log_{10}(4\pi h_t f/c)$. The second basis is

$$\mathrm{PL}_{\mathrm{A2G,NLoS}} = \lambda_0 - 16.16 + 12.0436\exp\left[-\frac{90-\theta}{7.52}\right]. \quad (10)$$

The conservative raw estimate is

$$P_0 = \mathrm{clip}(\max\{\mathrm{PL}_C, \mathrm{PL}_{\mathrm{A2G,NLoS}}\}, 0, 185). \quad (11)$$

The clipping is applied *before* P_0 and P_0^2 enter the regression.

The visibility mask identifies that a direct path is blocked but not which street canyon, roof edge, or clutter configuration dominates the remaining path. We therefore augment P_0 with exactly defined local quantities. Let $B(p) = \mathbf{1}\{T(p) \neq 0\}$ be the building indicator from topology height T; let $G = 1 - B$; and let $N(p) = \mathbf{1}\{m_{\mathrm{LoS}}(p) \leq 0.5\}G(p)$ be the NLoS-ground indicator. For odd k, define the reflect-padded box mean

$$\mathcal{A}_k[z](p) = \frac{1}{k^2} \sum_{q \in \mathcal{W}_k(p)} z(q). \quad (12)$$

The implemented features are

$$\begin{aligned} \ell_d &= \ln(1 + d_{2D}/1\,\mathrm{m}), & \delta_k &= \mathcal{A}_k[B], \\ h_k &= \mathcal{A}_k[TB]/90\,\mathrm{m}, & n_k &= \mathcal{A}_k[N], \\ \sigma_{\mathrm{sh}} &= 2.3197[\max(90-\theta, 0)]^{0.2361}, & & \\ \theta_n &= \mathrm{clip}(\theta/90, 0, 1), & k &\in \{15, 41\}. \end{aligned} \quad (13)$$

Thus, h_k is the window-averaged building height divided by 90 m, including ground pixels as zeros; it is not the mean conditioned only on buildings. The full vector, in implementation order, is

$$\begin{aligned} \mathbf{x} = [&P_0^2, P_0, \ell_d, \delta_{15}, \delta_{41}, h_{15}, h_{41}, \delta_{41}\ell_d, \\ &n_{15}, n_{41}, n_{41}\ell_d, \sigma_{\mathrm{sh}}, \theta_n, n_{41}\theta_n, 1]^\top. \end{aligned} \quad (14)$$

Table II removes any remaining ambiguity by giving the array operation, scale, and interpretation of every regressor. Window sizes are in pixels and therefore metres here; $\mathcal{A}_k$ always includes all k^2 cells after reflect padding, including zeros over ground in the height features. No standardization or centering is applied before the dot product.

The regression regime is observable. From a complete topology map, let D be its building-pixel fraction and H its mean nonzero building height. The class is dense/high-rise if $D \geq 0.2549$ or $H \geq 15.95\,\mathrm{m}$; open/low-rise if $D \leq 0.1957$ and $H \leq 10.91\,\mathrm{m}$; and mixed/mid-rise otherwise. Transmitter-height bins are low for $h_{\mathrm{tx}} \leq 58.12\,\mathrm{m}$, middle up to 103.85 m, and high above it. For regime q, a multiple linear (multilinear) regression on calibration pixels gives

$$\begin{aligned} \hat{\mathbf{w}}_q &= \arg\min_{\mathbf{w}} \sum_{i \in q} (\mathbf{x}_i^\top \mathbf{w} - y_i)^2, \\ \mathrm{PL}_{\mathrm{NLoS}} &= \mathrm{clip}(\mathbf{x}^\top \hat{\mathbf{w}}_q, 20, 185). \end{aligned} \quad (15)$$

The current artifact contains all 3×3 NLoS topology/height vectors; the inference code checks compatible height-bin fallbacks only if an exact key is absent. This branch is semi-empirical, but its raw estimate, each feature, selected regime, coefficient, and pre-clipping contribution remain auditable.

Table III gives selected exact coefficients for the middle height bin. Coefficients cannot be compared by magnitude alone because the features have different scales. Table IV therefore reports pixel-weighted feature values and products $w_j x_j$ on a deterministic audit of every tenth final-test map: 259 unseen maps and 3031586 NLoS pixels. The largest terms strongly cancel. In particular, the intercept, P_0, and P_0^2 average approximately $+179.7$, -132.6, and $+71.8$ dB. This makes the result sensitive to correlated polynomial terms and impairs interpretability. The next calibration iteration should remove P_0^2 and refit every regime; it must not simply drop the term while retaining the present coefficients.

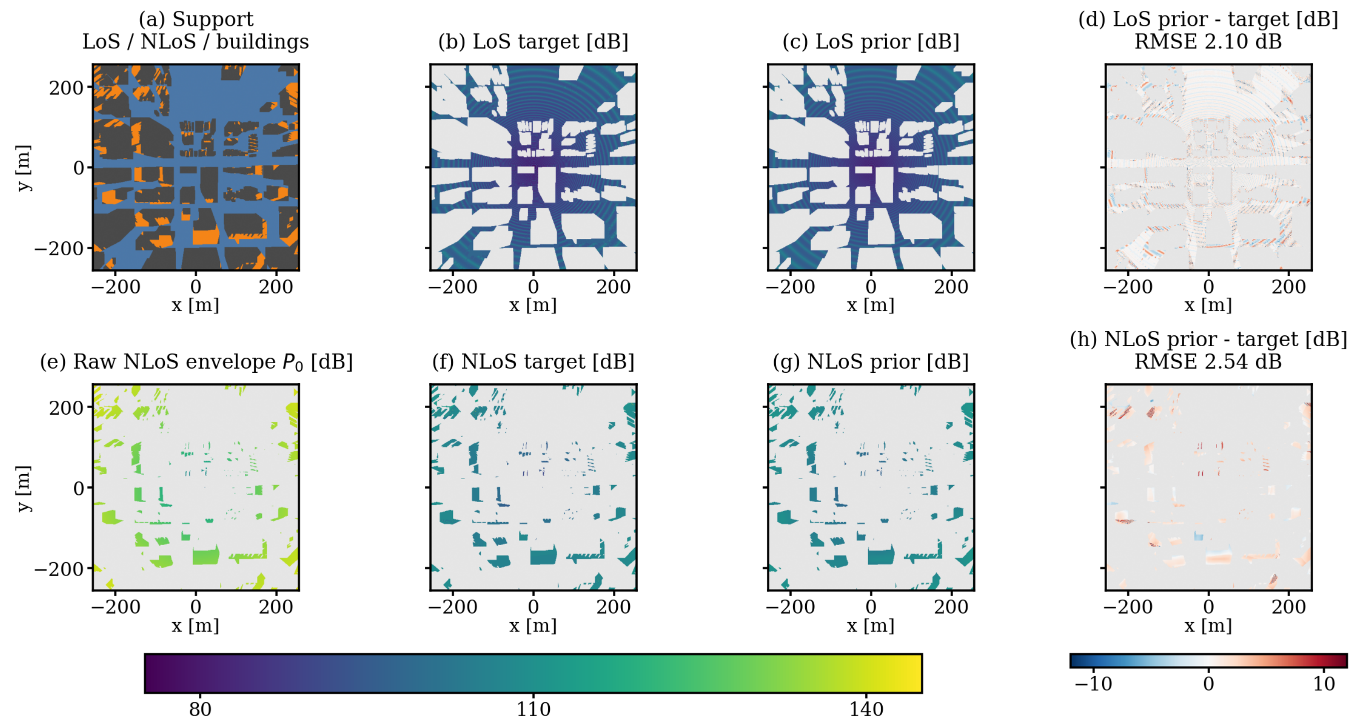

Fig. 2. A held-out example from Vancouver (“sample_15262”, $h_{\mathrm{tx}} = 50.30\,\mathrm{m}$, dense/high-rise, low-transmitter regime). Panel (a) shows the geometric support; (b)–(d) are the LoS target, frozen lookup-table prior, and signed error; (e) visualizes the raw physical NLoS envelope P_0; and (f)–(h) give the NLoS target, calibrated prior, and signed error. Grey excludes pixels outside the displayed regime or without a valid target. The unseen map contains 119180 LoS and 25607 NLoS receivers, with 2.10 dB and 2.54 dB RMSE, respectively.

TABLE II

EXACT CONSTRUCTION OF EVERY NLOS MULTILINEAR-REGRESSION INPUT AT RECEIVER CELL p. THE FEATURE ORDER IS ALSO THE COEFFICIENT-VECTOR ORDER. “UNITLESS” MEANS THAT THE STATED METRE OR DEGREE NORMALIZATION HAS ALREADY BEEN APPLIED.

j	Feature $x_j(p)$	Exact computation	Scale and role
1	P_0^2	$[\mathrm{clip}(\max\{\mathrm{PL}_C, \mathrm{PL}_{\mathrm{A2G,NLoS}}\}, 0, 185)]^2$	dB^2; quadratic raw trend
2	P_0	Eq. (11), after clipping	dB; raw urban/A2G envelope
3	ℓ_d	$\ln[1 + d_{2D}(p)/1\,\mathrm{m}]$	unitless; radial growth
4	δ_{15}	$15^{-2} \sum_{q\in\mathcal{W}_{15}(p)} \mathbf{1}\{T(q) \neq 0\}$	unitless; local building fraction
5	δ_{41}	$41^{-2} \sum_{q\in\mathcal{W}_{41}(p)} \mathbf{1}\{T(q) \neq 0\}$	unitless; contextual building fraction
6	h_{15}	$(15^2\,90\,\mathrm{m})^{-1} \sum_{q\in\mathcal{W}_{15}(p)} T(q)\mathbf{1}\{T(q) \neq 0\}$	unitless; local height mass
7	h_{41}	$(41^2\,90\,\mathrm{m})^{-1} \sum_{q\in\mathcal{W}_{41}(p)} T(q)\mathbf{1}\{T(q) \neq 0\}$	unitless; contextual height mass
8	$\delta_{41}\ell_d$	$x_5(p)x_3(p)$	unitless; density–range interaction
9	n_{15}	$15^{-2} \sum_{q\in\mathcal{W}_{15}(p)} \mathbf{1}\{m_{\mathrm{LoS}}(q) \leq 0.5\}\mathbf{1}\{T(q) = 0\}$	unitless; local blocked-ground fraction
10	n_{41}	$41^{-2} \sum_{q\in\mathcal{W}_{41}(p)} \mathbf{1}\{m_{\mathrm{LoS}}(q) \leq 0.5\}\mathbf{1}\{T(q) = 0\}$	unitless; contextual blocked-ground fraction
11	$n_{41}\ell_d$	$x_{10}(p)x_3(p)$	unitless; blockage–range interaction
12	σ_{sh}	$2.3197[\max\{90 - \theta(p), 0\}]^{0.2361}$	dB; elevation-dependent shadowing proxy
13	θ_n	$\mathrm{clip}[\theta(p)/90, 0, 1]$	unitless; normalized elevation angle
14	$n_{41}\theta_n$	$x_{10}(p)x_{13}(p)$	unitless; blockage–elevation interaction
15	1	an all-ones array with the same 513×513 shape	unitless; intercept

The complete prior uses the deterministic LoS mask m_{LoS}:

$$\mathrm{PL}_{\mathrm{prior}} = m_{\mathrm{LoS}}\mathrm{PL}_{\mathrm{LoS}} + (1 - m_{\mathrm{LoS}})\mathrm{PL}_{\mathrm{NLoS}}. \tag{16}$$

IV. RESULTS

Table V gives the final attenuation-only test result. The prior reaches 1.9383 dB RMSE and 1.2319 dB MAE across all valid receiver pixels. The map correlation of 0.947710 shows that its accuracy is not obtained only by matching a global offset: relative high- and low-attenuation regions are generally placed correctly. The LoS error is 1.7519 dB. NLoS remains the hard regime at 3.5143 dB, consistent with the unobserved identity of the dominant diffracted or reflected path after direct-path blockage.

Height is part of the estimator rather than a post-hoc reporting variable. Table VI shows the frozen prior over held-out validation and test cities for four physical height intervals. Error decreases from 2.180 dB at 12 m to 50 m to 1.627 dB above 300 m. The largest change is in NLoS, where low transmitters interact more strongly with local rooftops

TABLE III
EXAMPLES OF EXACT NLOS MULTILINEAR-REGRESSION COEFFICIENTS FOR THE MIDDLE TRANSMITTER-HEIGHT BIN. THE COMPLETE 9×15 COEFFICIENT TABLE IS AVAILABLE IN THE GITHUB PAPER REPOSITORY.

Feature	Open/low-rise	Mixed/mid-rise	Dense/high-rise
P_0^2	0.001605	0.003632	0.005737
P_0	-0.500636	-0.994251	-1.409844
ℓ_d	8.167217	6.663205	4.112897
δ_{41}	-14.937161	-28.582537	-34.561308
n_{41}	12.112861	16.509693	12.404412
σ_{sh}	-8.316422	-2.183255	-3.674099
θ_n	-17.901197	-4.058654	-7.714892
Bias	157.615565	153.694717	195.570915

TABLE IV
LARGEST MEAN NLOS REGRESSION TERMS ACROSS THE 259-MAP HELD-OUT AUDIT. $\bar{w}$ IS THE NLOS-PIXEL-WEIGHTED APPLIED COEFFICIENT, $\bar{x}$ IS THE MEAN FEATURE VALUE, AND THE LAST COLUMNS ARE PRE-CLIPPING CONTRIBUTIONS IN DB. THESE ARE SCALE DIAGNOSTICS, NOT CAUSAL ABLATIONS.

Feature	$\bar{w}$	$\bar{x}$	$\overline{wx}$	$\overline{\lvert wx \rvert}$
Bias	179.741	1.000	179.741	179.741
P_0	-1.010	131.273	-132.611	134.396
P_0^2	0.004158	17272.148	71.847	72.366
σ_{sh}	-5.504	6.299	-34.739	34.739
ℓ_d	4.634	5.263	24.418	24.418
δ_{41}	-31.006	0.212	-6.371	6.372
n_{41}	10.135	0.578	5.534	5.534
$\delta_{41}\ell_d$	4.620	1.105	4.947	4.954

TABLE V
FROZEN PRIOR ON THE 2590-MAP, 14-CITY TEST SET.

Metric	Value	Unit
Overall RMSE	1.9383	dB
Overall MAE	1.2319	dB
LoS RMSE	1.7519	dB
NLoS RMSE	3.5143	dB
Map correlation	0.947710	–

TABLE VI
PRIOR RMSE BY HEIGHT ON HELD-OUT VALIDATION AND TEST CITIES.

Height	Overall	LoS	NLoS	Maps
12–50 m	2.180	1.797	3.756	1211
50–150 m	1.936	1.785	3.228	2634
150–300 m	1.680	1.666	2.259	786
300–500 m	1.627	1.619	2.255	362

and street canyons. The trend does not imply that altitude alone solves dense prediction; it shows why a single height-independent calibration would combine physically different regimes.

The result is best interpreted as a strong physical/statistical anchor rather than a universal propagation model. The coefficients are tied to the fixed frequency, antenna assumptions, simulator, and available visibility mask. The city split tests transfer across urban layouts, but the labels still come from ray tracing rather than field measurements.

A. Flat-Terrain Limitation and Future Work

All current scenes use flat terrain. Consequently, the reported visibility, distance, and height relationships do not account for terrain-induced blockage or for changes in local ground elevation. Slopes and ridges can alter both the LoS boundary and the meaning of transmitter height relative to nearby receivers. Follow-up work will introduce non-flat digital terrain, compute antenna heights and reflected-path geometry relative to local elevation, add terrain descriptors to the NLoS calibration, and evaluate transfer across relief classes. Field measurements are also required to separate simulator fit from over-the-air validity.

V. CONCLUSION

This paper presented an attenuation prior for dense, variable-height UAV CKMs. The prior separates a coherent, height-calibrated LoS model from a morphology-calibrated NLoS model and freezes every fitted coefficient before unseen cities are evaluated. On 2590 maps from 14 held-out cities, it reaches 1.9383 dB overall RMSE and 0.947710 map correlation. Its main practical value is transparency: stable range, height, visibility, and morphology structure is captured before a more flexible model is considered. The next steps are an explicit component ablation, non-flat terrain, and measurement validation.

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